Absolute Illuminance Analysis — Bright Sunny Afternoon

Does Array Attenuation Cross a Functional-Vision Threshold?

IYPT Pinhole Sunglasses — Array Geometry Sub-Study

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1 Why Absolute Illuminance, Not Just % Transmission

Percentage transmission alone cannot say whether a wearer can actually see. A design transmitting 1% of incident light could be perfectly usable in bright sun (where 1% of a very large number is still a lot of light) or dangerously dark indoors (where 1% of a small number is almost nothing). The physically meaningful question is the *absolute* illuminance reaching the eye, compared against known functional-vision thresholds.

2 Method

For each winning design, best-case transmission T (single active hole, fully centred in the pupil) was computed as $T = \pi(a_{\text{hole}}/2)^2/\pi R_p^2$, with $R_p = 2000\,\mu$m (4 mm pupil). Illuminance reaching the eye is $E_{\text{eye}} = T \times E_{\text{ambient}}$, evaluated at three standard reference ambient levels (IES/CIE lighting handbook values): direct summer sunlight (100,000 lux), general bright-sun scene illuminance (32,000 lux), and an overcast day (5,000 lux, used as a cross-check for variable conditions).

3 Results

M (D)	a_{hole} (μm)	T (%)	Direct sun (100k lux)	Bright day (32k lux)	Overcast (5k lux)
0	2000	25.00	25,000	8,000	1,250
1	1250	9.77	9,766	3,125	488
2–3	750	3.52	3,516	1,125	176
4–5	600	2.25	2,250	720	112
6	500	1.56	1,562	500	78
8	**400**	**1.00**	**1,000**	**320**	**50**
global	2000	25.00	25,000	8,000	1,250

Reference functional-vision thresholds: minimum navigable (twilight) ≈10 lux; minimum comfortable reading ≈100 lux; typical comfortable office lighting ≈500 lux; bright overcast day ≈5,000 lux.

Key Result

Under direct-sun ambient (100,000 lux), **every design stays well above all functional-vision thresholds**. Even the darkest design (M=8, 1% transmission) delivers 1,000 lux to the eye — comparable to a bright overcast day, 2× above typical office-comfortable lighting, and 100× above the minimum navigable threshold. No design impairs vision in its intended bright-sun use case.

4 The Variable-Lighting Caveat

This result answers the specific question asked — performance in a bright sunny afternoon — and the answer is unambiguous: no threshold is crossed, with large margin. A separate consideration exists for lighting conditions *other* than bright sun, since sunglasses are not only worn in direct sunlight.

Caveat

Under overcast ambient (5,000 lux, a common daytime condition), the M=8 design delivers only ≈50 lux — approaching the low end of comfortable functional vision. This is not a flaw specific to this array; it is the standard reason general-purpose sunglass standards (e.g. EN ISO 12312-1) cap how dark a lens may be: not because bright-sun performance is inadequate, but because a lens dark enough for full sun becomes marginal once ambient light drops. This is a known, general trade-off of any fixed-attenuation eyewear, not a defect unique to the pinhole array design.

5 Conclusion

The array's attenuation, evaluated against the actual physical quantity that determines whether a wearer can see (absolute illuminance, not percentage transmission), satisfies the problem statement's light-reduction requirement (R2) without crossing any functional-vision threshold in bright sunlight — the design's intended operating condition. The one open consideration, honestly stated rather than omitted, is reduced margin under lower ambient conditions at the highest myopia level (M=8), consistent with a general property of high-attenuation eyewear rather than a specific failure of this design.